

Must Read Topics for Interviews

1. OOPS

Four Pillars (Encapsulation, Abstraction, Inheritance, Polymorphism)

Runtime vs Compile Time Polymorphism

Method Overloading vs Overriding

Interface vs Abstract Class

Association, Aggregation, Composition

SOLID Principles

Constructor Chaining

Object Cloning (Deep vs Shallow Copy)

Static vs Instance Members

Access Modifiers

Diamond Problem

Dependency Injection

Cohesion vs Coupling

Must Read Topics for Interviews

Immutable Objects

Design Patterns (Singleton, Factory, Strategy)

Resources

[Refactoring Guru](#)

Java Brains

Concept && Coding

CodeWithHarry

2. DBMS

ACID Properties

Normalization (1NF-BCNF)

Denormalization

Primary vs Foreign Key

Indexing

Clustered vs Non-Clustered Index

Must Read Topics for Interviews

B+ Trees

Joins (All Types)

Transactions

Locks (Shared/Exclusive)

Deadlock

Isolation Levels

CAP Theorem

Sharding

SQL vs NoSQL

Resources

[GeeksforGeeks DBMS](#)

Gate Smashers

Neso Academy

3. Computer Networks

Must Read Topics for Interviews

TCP vs UDP

OSI Model

TCP/IP Model

Three-Way Handshake

Four-Way Connection Termination

HTTP vs HTTPS

DNS Working

Load Balancer

Reverse Proxy

SSL/TLS

Cookies vs Sessions

REST APIs

WebSocket

CDN

How Browser Opens Google.com

Must Read Topics for Interviews

Resources

Neso Academy

ByteByteGo

Computer Networking Full Course FreeCodeCamp

4. Operating System

Process vs Thread

Context Switching

Multithreading

Scheduling Algorithms

Deadlock

Starvation

Race Condition

Mutex

Semaphore

Must Read Topics for Interviews

Paging

Segmentation

Virtual Memory

Heap vs Stack

Memory Allocation

Garbage Collection

Resources

Neso Academy

Gate Smashers

[Operating Systems Three Easy Pieces](#)

5. Puzzles

25 Horses Puzzle

8 Ball Puzzle

Bridge and Torch

Must Read Topics for Interviews

Water Jug

100 Prisoners

Egg Dropping

Coin Flip

Poisoned Bottle

Pirate Gold Division

Clock Angle

Train Crossing

River Crossing

Probability Dice Questions

Monty Hall Problem

Logical Deduction Puzzles

Resources

InterviewBit Puzzles

[GeeksforGeeks Puzzles](#)

Must Read Topics for Interviews

6. Linux

What happens when you type a command in terminal?

Difference between Process and Thread?

What is a Zombie Process?

What is a Daemon Process?

Difference between Hard Link and Soft Link?

What is an Inode in Linux?

Explain `chmod 755`.

Difference between `kill` and `kill -9`.

Difference between `grep`, `find`, and `locate`.

Difference between `>` and `>>`.

What is Pipe `|` in Linux?

What is Swap Memory?

What happens when RAM gets full?

Must Read Topics for Interviews

Difference between TCP and UDP?

Difference between HTTP and HTTPS?

What is SSH?

What is Cron Job?

What is Environment Variable and PATH?

Explain `fork()` and `exec()`.

Your production server is down — how will you debug?

Resources

TechWorld with Nana

[Linux Journey](#)

[The Linux Command Line Guide](#)

7. Git

What is Git?

Difference between Git and GitHub?

Must Read Topics for Interviews

What is a Repository?

Difference between `git pull` and `git fetch`?

Difference between `git merge` and `git rebase`?

What is a Branch in Git?

Why do we use Pull Requests?

What is Git Stash?

Difference between `git reset` and `git revert`?

What is Merge Conflict?

How do you resolve merge conflicts?

Difference between Local Repository and Remote Repository?

What is `.gitignore`?

Difference between `git clone` and `git fork`?

What is HEAD in Git?

Difference between `git add .` and `git add <file>`?

Must Read Topics for Interviews

What is Detached HEAD state?

Explain Git workflow in teams.

How do you undo a commit?

How do you debug when code works locally but fails after merge?

Resources

[Git Documentation](#)

CodeWithHarry Git Playlist

Kunal Kushwaha Git Playlist

8. SQL

Find nth highest salary using SQL

Find the 2nd highest salary from Employee table

Find duplicate records in a table

Delete duplicate rows from a table

Find highest salary department-wise

Must Read Topics for Interviews

Find employees earning more than department average

Find employees with same salary

Find manager name along with employee name using self join

Find departments having more than 5 employees

Find customers who never placed orders

Difference between DELETE, DROP, and TRUNCATE

Difference between WHERE and HAVING

Difference between INNER JOIN and LEFT JOIN

What is normalization and its types?

Difference between PRIMARY KEY and FOREIGN KEY

What is indexing and why is it used?

Difference between UNION and UNION ALL

What are ACID properties in SQL?

Difference between clustered and non-clustered index

What is subquery vs correlated subquery?

Must Read Topics for Interviews

Resources

[LeetCode SQL 50](#)

[Mode SQL Tutorial](#)

[StrataScratch](#)

9. Language Core (Java/C++/Python)

C++ — Top 10 Must Asked Core Questions

Stack vs Heap memory

Difference between compile time and runtime polymorphism

What happens during compilation in C++?

What is virtual function and how vtable works?

Difference between shallow copy and deep copy

Copy constructor vs move constructor

What are smart pointers and why needed?

Difference between process and thread

What causes memory leak and segmentation fault?

Must Read Topics for Interviews

Difference between malloc/free and new/delete

Java — Top 10 Must Asked Core Questions

Difference between JDK, JRE, JVM

How Java achieves platform independence?

Heap vs Stack memory in JVM

How Garbage Collection works in Java?

Difference between == and equals()

Difference between HashMap and ConcurrentHashMap

Process vs Thread in Java

What is synchronization and deadlock?

What happens when a Java program runs internally?

Difference between String, StringBuilder, and StringBuffer

Python — Top 10 Must Asked Core Questions

What is GIL (Global Interpreter Lock)?

Must Read Topics for Interviews

Difference between multithreading and multiprocessing

Mutable vs Immutable objects

Difference between is and ==

Deep copy vs shallow copy

How Python memory management works?

What happens when Python code executes internally?

List vs Tuple

What are generators and why are they memory efficient?

Difference between interpreter and compiler in Python

Resources

Java Brains

Coding with John

Kunal Kushwaha

Corey Schafer

Must Read Topics for Interviews

10. Projects / Behavioral

Explain Your Project Architecture

Biggest Technical Challenge

Why This Tech Stack?

Scalability Improvements

Database Design Decisions

Security Implementations

Performance Optimizations

CI/CD Pipeline

Monitoring & Logging

Caching Strategy

Failure Handling

Team Conflict Situation

Leadership Example

Production Bug You Solved

Must Read Topics for Interviews

Why Should We Hire You?

Resources

[Exponent Behavioral Interviews](#)

[Amazon Leadership Principles](#)

ByteByteGo

Resources

DBMS

- GeeksforGeeks DBMS — <https://www.geeksforgeeks.org/dbms/>
- Gate Smashers — <https://www.youtube.com/@GateSmashers>
- Neso Academy — <https://www.youtube.com/@nesoacademy>

OOPS

Must Read Topics for Interviews

- Refactoring Guru — <https://refactoring.guru/>
- Java Brains — <https://www.youtube.com/@Java.Brains>
- Concept & Coding — <https://www.youtube.com/@ConceptandCoding>
- CodeWithHarry — <https://www.youtube.com/@CodeWithHarry>

Computer Networks

- Neso Academy — <https://www.youtube.com/@nesoacademy>
- ByteByteGo — <https://www.youtube.com/@ByteByteGo>
- FreeCodeCamp CN Course — <https://www.youtube.com/@freecodecamp>

Operating System

- Neso Academy — <https://www.youtube.com/@nesoacademy>
- Gate Smashers — <https://www.youtube.com/@GateSmashers>
- Operating Systems: Three Easy Pieces — <https://pages.cs.wisc.edu/~remzi/OSTEP/>

Language Core (Java / C++ / Python)

- Java Brains — <https://www.youtube.com/@Java.Brains>
- Coding with John — <https://www.youtube.com/@CodingWithJohn>
- Kunal Kushwaha — <https://www.youtube.com/@KunalKushwaha>
- Corey Schafer — <https://www.youtube.com/@coreyms>

Puzzles

- InterviewBit Puzzles — <https://www.interviewbit.com/puzzles/>
- GeeksforGeeks Puzzles — <https://www.geeksforgeeks.org/puzzles/>

Linux

- TechWorld with Nana — <https://www.youtube.com/@TechWorldwithNana>
- Linux Journey — <https://linuxjourney.com/>
- The Linux Command Line Guide — <https://linuxcommand.org/>

Git

Must Read Topics for Interviews

- Git Documentation — <https://git-scm.com/doc>
- CodeWithHarry Git Playlist — <https://www.youtube.com/@CodeWithHarry>
- Kunal Kushwaha Git Playlist — <https://www.youtube.com/@KunalKushwaha>

SQL

- LeetCode SQL 50 — <https://leetcode.com/studyplan/top-sql-50/>
- Mode SQL Tutorial — <https://mode.com/sql-tutorial/>
- StrataScratch — <https://www.stratascratch.com/>

Projects & Behavioral

- Exponent Behavioral Interviews — <https://www.tryexponent.com/>
- Amazon Leadership Principles — <https://www.amazon.jobs/content/en/our-workplace/leadership-principles>
- ByteByteGo — <https://www.youtube.com/@ByteByteGo>